

# Forest Protection

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## Science

May, 2026



**Short Title:** Forest Protection  
**Faculty** Science and Computing  
**Department** Science  
**Cluster** Forestry  
**Author** NMCCARTHY  
**Credits** 5

**Level:** Introductory

## Description of Module / Aims

This module will help students to obtain knowledge of the concepts and techniques for understanding, identifying and controlling tree diseases, disorders and pests. It will also give them the ability and confidence to diagnose problems in plantation forestry, and suggest remedies for these. Students will also learn about the problems associated with fire and wind in plantations, and assess ways of reducing these problems.

## Programmes

FORS-0008 BSc in Forestry (WD\_SFORS\_D)

2 / 3 / M

## Indicative Content

- Introduction to forest pathology
- Abiotic agents: frost, chemicals, pollution
- Mammal and bird pests and their control
- Wind and windthrow hazard classification
- Important insect pests and their control
- Important forest diseases, their life cycles and control
- Fire: ecology of fire; losses and susceptibility of forests to fire; nature and prevention of forest fires

## Learning Outcomes

*On successful completion of this module, a student will be able to:*

1. Describe the abiotic agents causing problems in Irish forests.
2. Identify and recognise the symptoms of the important forest pests and diseases.
3. Record and collect both insect and foliage samples for dispatch to a laboratory.
4. Use the windthrow hazard class system.
5. Describe the implications of fire in forests and outline methods of reducing the incidence of outbreaks.
6. Describe the life cycles and control of forest pests and diseases.
7. Review relevant papers in the field of forest protection.

## Learning and Teaching Methods

- Lectures.
- Seminars.
- Field trips.
- Self learning.
- Demonstrations.
- Field work.
- Reports.

## Assessment Methods

|                           | Weighing | Outcomes Assessed |
|---------------------------|----------|-------------------|
| Final Written Examination | 60%      | 1,5,6             |
| Continuous Assessment     | 40%      |                   |
| <i>Practical</i>          | 30%      | 2,3,4             |
| <i>Case Studies</i>       | 10%      | 7                 |

## Assessment Criteria

- <40%:** Very limited knowledge and understanding of the subject matter. Lacking in technical ability/skill and understanding to carry out assessments. Failure to meet the objectives of assessments. Unable to carry out or submit independent study and research. Unable to present written work with attention to correct formatting, grammar and spelling.
- 40%-49%:** Demonstrate a limited knowledge of the subject matter. Show some technical ability/skill in carrying out and reporting on assessments. Attempt made to meet objectives and carry out independent study and research. Final result lacks balance and arguments undeveloped. Can present written work correctly formatted with minor grammar and spelling errors only.
- 50%-59%:** Demonstrate satisfactory general knowledge of the main issues within the subject. Logically and competently carry out assessments. Satisfactory handling and awareness in carrying out independent study and research. Reporting and projects adequately structured and referenced.
- 60%-69%:** Demonstrate sound knowledge of the subject matter. Show ability to analyse and logically argue in an effective and mature style. Demonstrate initiative and the ability to criticise methods used and appreciation of experimental design and current knowledge limitations when carrying out assessments. Projects and reporting coherent and soundly structured illustrating wide and in depth reading on subject coupled with the proper use of references.
- 70%-100%:** Demonstrate authoritative handling of complex material and a highly developed and extensive understanding of the subject. Provide well focused analysis and convincing argument on the subject. Demonstrate imaginative approach, excellent technical ability, awareness, and aptitude in carrying out and reporting on assessments. Show initiative, individual thought, and enthusiasm in self study and independent learning.

### Note:

- In addition to the normal regulations, a minimum of 35% must be achieved in both the final exam and continuous assessment components of the module. In cases where continuous assessment has a practical component attendance of at least 75% in the practicals is mandatory.

## Learning Modes

| Learning Mode        | F/T Hours | P/T Hours |
|----------------------|-----------|-----------|
| Lecture              | 24        |           |
| Field Trips          | 24        |           |
| Independent Learning | 87        |           |

## Essential Material

- Forest Service, . \_Forest protection guidelines\_. Ireland: Forest Service, 2002.
- Rooney, S. and T.J. Hayden. \_Forest Mammals - Management and Control\_. Ireland: COFORD, 2002.

- Strouts, R.G. and T.G. Winters. *\_Diagnosis of Ill-health in trees\_* . 2nd ed. UK: TSO, 2013.
- Willoughby, I., H. Evans and J. Gibbs. *\_Reducing pesticide use in forestry\_*. UK: Forestry Commission, 2004.

## **Supplementary Material**

- Blanchard, R.O. and T.A. Tattar. *\_Field and laboratory guide to tree pathology\_*. UK: Academic Press Inc, 1981.
- Gregory, S.C. and D.B. Redfern. *\_Diseases and disorders of forest trees\_*. UK: TSO, 1998.
- Springthorpe, G.D. and N.G. Myhill. *\_Wildlife Rangers handbook (Forestry Commission handbook 10)\_*. UK: TSO, 1994.

## **Requested Resources**

- Lecture Room: Loose Seated